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Optimization of Water Injection in a Giant Carbonate Reservoir: Lessons, Challenges and Solutions

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Abstract

Water injection is used to improve the overall efficiency of hydrocarbon recovery while prolonging the productive life of the reservoir. It is imperative to have an efficient and optimized water injection system to maximize the value of water injected. This paper summarizes the challenges, learnings, and short- and long-term solutions for optimizing the water injection system in a giant, saturated carbonate reservoir in the Middle East which has been in production for over 60 years.

Overhauling the existing decades old system is difficult and needs a coordinated team effort. This paper discusses a multi-pronged effort for water injection optimization. A robust reservoir surveillance plan coupled with in-house AI-ML based tools helps to evaluate performance and identify issues. System inefficiencies were identified proactively through regular inspection and real time data. To arrest the reservoir pressure depletion, production rebalancing was envisaged to increase production from healthy areas. Water injection optimization along with ramp up water injection volumes was undertaken. MRC (maximum reservoir contact) wells and lower completion were proposed to reduce surface congestion while improving accessibility and conformance.

Several short- and long-term solutions have been developed which are in different phases of implementation. The field suffers from a lack of water required for injection resulting in a gradual drop in reservoir pressure. Also, the increased withdrawal is causing gas cap gas cusping in attic wells, thus constraining surface facilities. Employing conventional and AI-based analysis tools, efforts have been made to increase the value of injected water while limiting the reservoir pressure drop by identifying the right locations and correct quantities for injection. Simplified clusters with low turn-around time along with few conventional clusters have helped to add significant volumes. Proactive stimulation, production-injection rebalancing, MRC wells, smart completions and innovative artificial lift ideas have helped in arresting pressure decline, improving well accessibility for logging, and reducing carbon footprint. Over the long-term, seawater will replace the current aquifer water injection which will help in increasing injection capacity. Injection along the gas-cap fence is under implementation to minimize gas migration to oil zone. These initiatives are expected to add significant reserves and achieve ~60% RF by the end of the concession.

The current paper showcases the need for adopting a multipronged strategy covering all aspects of water injection management for unlocking potential while maintaining reservoir health of a brownfield. The

lessons learned and the improvised solutions utilized in the field can be utilized for other water-deficient fields for maximizing the value of injected water while minimizing pressure loss in the short term while deploying long term solutions for improving water injection volumes.

Introduction

The field is a giant asymmetrical anticlinal structural configuration formed in the early Cretaceous. It has several stacked oil and gas bearing reservoirs. The target, reservoir-B, is one of these saturated carbonate reservoirs. The Upper unit of the reservoir is heterogeneous and exhibits better reservoir quality due to the deposition of higher energy facies. The Lower zone is more homogenous and is relatively poorer in quality due to the presence of muddier depositional facies. The STOIIP is split in the ratio of 40:60 in Upper and Lower zones respectively and the average permeability of the Upper zone is about 10 to 100 times that of Lower zone (Figure 1). The reservoir has weak aquifer support and a gas cap. The reservoir development is supported by a combination of peripheral water injection, pattern line drive water injection and lean gas injection into the gas cap. Field performance shows that the water injected in Lower zone tends to crossflow to Upper zone (Figure 2) and therefore the Lower zone has remained largely unflooded.

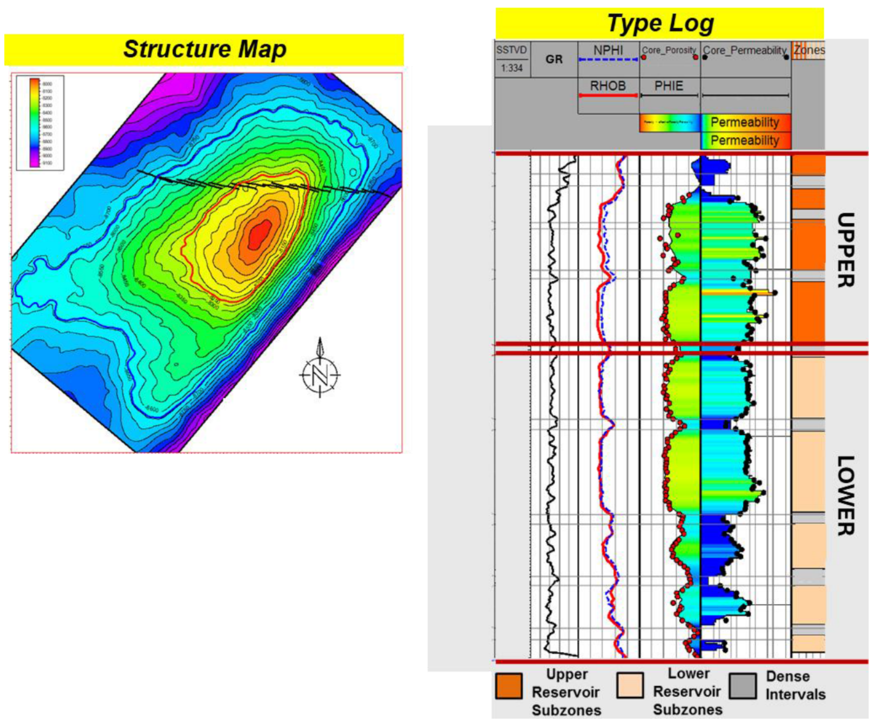

Figure 1—Typical Petrophysical Log for the Reservoir-B

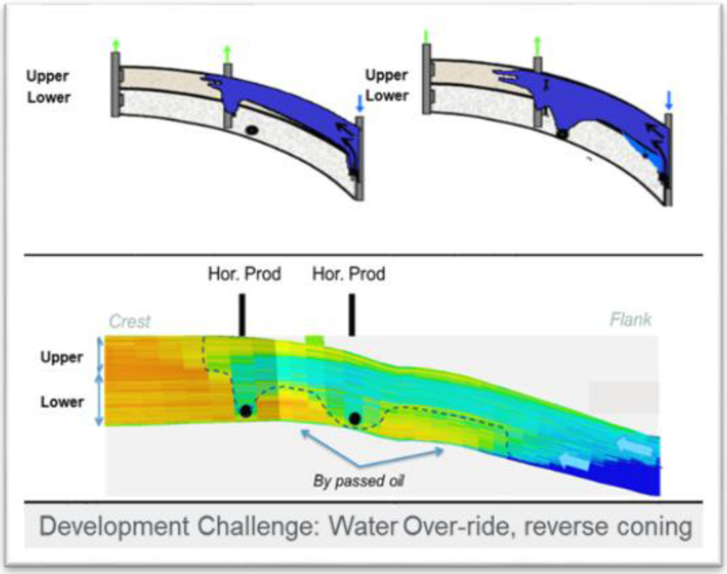

Figure 2—Water over-ride from Lower to Upper Zone leaves bypassed oil

The reservoir temperature and pressure are 250° F and 4000 psi respectively. The formation brine has a salinity of about 180,000 ppm and the produced oil API gravity is about 41°. The carbonate reservoir is slightly oil wet. A vast amount of pressure, PVT, SCAL and time lapse RST data are available.

The field started production in the 1960s and the water injection was started a decade later in the 1970s. The production and injection system for the field have been designed to handle the production and injection requirements for all the reservoirs using the same surface facilities. As with most reservoirs and fields, with acquisition of more and more data and technological advancements, higher production potential was identified. With a new development strategy, the withdrawal and injection rates for the field and the reservoir-B were increased from the 90s. The existing surface facilities were modified and revamped to handle more production. The injection facilities were also added and improved to handle increased voidage but have been largely inadequate to meet the ideal overall VRR requirement of 1. Considering the gas cap expansion and the lean gas injection in the gas cap, a VRR of 0.8 is considered to be sufficient for maintaining good reservoir health.

However, the distribution of the water volumes is not the same in all the parts of the reservoir. Some regions are affected by lack of injection and this has resulted in development of low pressure sink areas in parts of the reservoir. The expansion of gas cap is causing the GOR to rise in the attic producers which in turn is constraining the surface production facilities. Gas cap fence injection is planned to limit gas cap expansion. However, the limited injection volumes are a hindrance to the implementation of reservoir management and field development plan. Over the long term, an ambitious project of sea-water injection is planned to replace the existing injection system which will provide ample amount of injection volumes for maintaining the reservoir health. But in the meantime, the existing infrastructure needs to be tweaked in order to limit or even remedy the deterioration of reservoir health while meeting the production requirements. This involves a multi-pronged strategy involving all the stakeholders - operations, drilling, projects and reservoir management teams - to work individually as well as together to identify issues and come up with a mitigation plan. This paper focuses on these short term initiatives in the water injection system for better reservoir management.

Operations

Water Injection System Overview

The water injection system in place is a cluster-based system. There are currently 41 water injection clusters (WIC) operational in the field. Each water injection cluster consists of the following:

1. A water supply well (WSW)
2. A water disposal well (WDW)
3. A surface injection pump (SIP)
4. Water injection wells (WIW) connected to each cluster (generally 8-10)

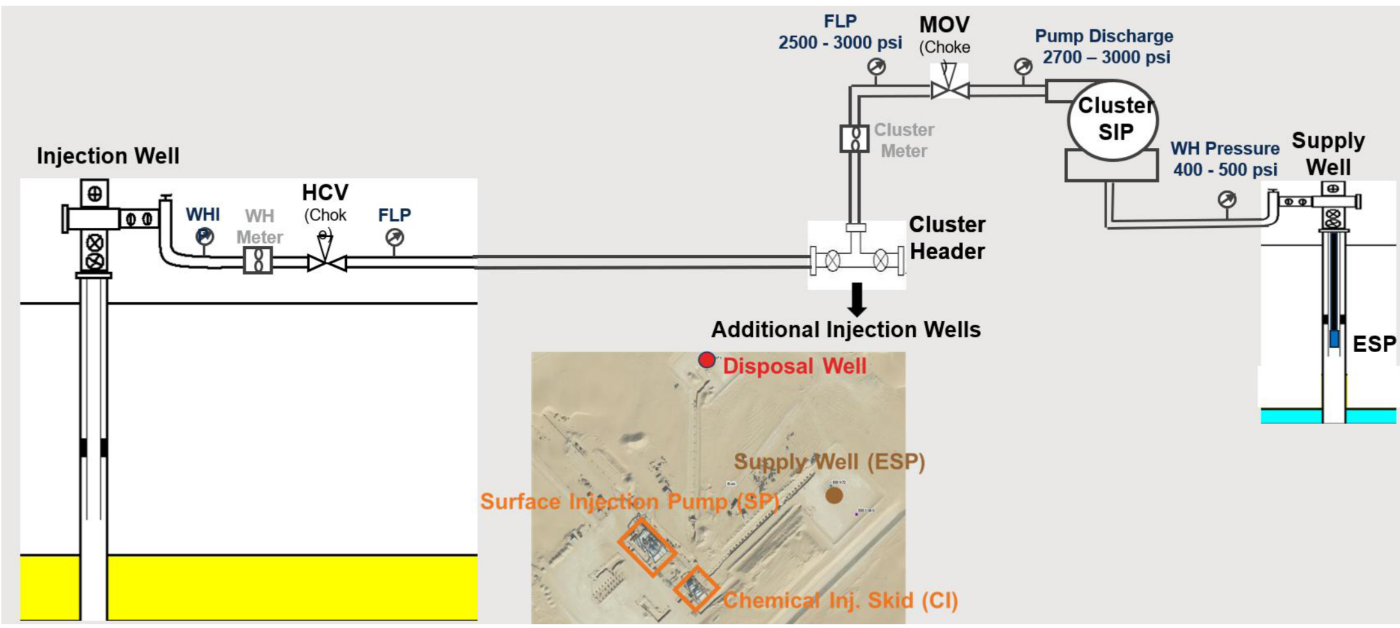

Figure 3—A typical water injection cluster layout

The WIC are shared by all the reservoirs. Each WIC has water injection capacity of 25,000 - 35,000 BWPD. The water injection volumes in a cluster are shared between the water injectors as per agreement between the reservoir management teams for each reservoir considering the injection capacity, reservoir requirements, WSW capacity and the injection efficiency to maximize the value of injection water at each cluster. Produced water re-injection (PWRI) is underway at 2 WICs. It is planned to be expanded to 2 more WICs with the start of sea-water injection.

In addition to the WICs, of late, emphasis has been placed on adding water injection capacity using Simplified Water Injection Clusters (SWIC). A SWIC is classified as a WI cluster where the water injection wells are directly connected to the discharge from the water supply well with no Surface Pump needed to boost the pressure. The capacity of each SWIC ranges between 20,000 - 25,000 bwpd. Each SWIC caters to ~3-5 WIWs.

With the current WICs and SWICs, the total injection capacity of the field is ~1270 MBD. However, due to the reasons that will be detailed later, the injection is maintained around 1000 MBD on average. This is insufficient with respect to meeting the VRR requirements with the current production requirements. Reservoir-B accounts for ~75% of total production from the field but is allocated ~65% of the total injection volume due to the inherent deficiencies of the current injection system.

The WIC and SWIC draw water from the underground saline aquifers. With the advent of sea-water injection, the clusters will be slowly phased out and the injection requirements will be met completely by the sea water and the PWRI. The sea-water injection project is expected to commence in a couple of years, increasing the capacity of the system to ~2000 MBD. However, to account for any potential delays in the sea-water injection project commencement, enhancements and capacity addition to the existing system are being planned and executed considering the overall value addition and commercial viability.

Challenges and Solutions

Drilling HSE Requirement. During drilling/workover of wells, the nearby injector wells within 1 km radius need to be either shut in or have the injection volumes reduced for HSE requirements. This has resulted in a loss of ~70 MBD injection capacity on average throughout the last year. Sometimes the entire cluster needs to be shut in due to inadequate injection volumes available.
Mitigation
Close liasoning with the RM teams has resulted in minimizing the shut in times for the wells by opening the wells as soon as drilling is finished. The surface level agreement to finalize the distance requirement for well shut-in during drilling is currently being reviewed through extensive data gathering and reservoir simulation studies.

Flowline leaks. The injection system has been in place since the 1970s. The ageing infrastructure has severe corrosion in some parts of the field which results in frequent leaks. In case of injection downtime, the corrosion is faster and hence, the longer the downtime, the higher the chance of flowline leaks. More than 300 cases of flowline leaks have been reported and rectified since 2020.
Mitigation
Several efforts are being undertaken to address the problem of flowline leaks. A leak reduction task force has been initiated to identify and rectify the flowline leaks at the earliest. State of the art quick leak repair methodology is being utilized. Efforts are also being made to identify a suitable corrosion inhibitor for the field to minimize corrosion. Reinforced Thermoplastic (RTP) pipelines are also being tested in few pilot wells. Over the long term, work is ongoing to replace all the existing flowlines before the implementation of seawater injection. Mothballing of flowlines is undertaken wherever the shut in duration is expected to be more than a quarter to avoid the corrosion due to standing water.

ESP and SIP seal Failures. The WSW ESPs and seal of the SIP undergo frequent failures. This results in significant downtimes for each of the clusters.
Mitigation
Close liaison with the site operations and ESP team has resulted in significant reduction in downtime for ESP and SIP seal replacement. A dedicated workover unit for ESP replacement in WSW is in place.

SWICs have been proposed and executed in the field to avoid the frequent surface pump maintenance related shutdowns. Till date, 6 SWICs have been commissioned and 9 additional SWICs are in the execution phase.

Real Time Monitoring and Manpower. As most of the systems are quite old, real-time monitoring of majority of the injection and production system is not available. This has led to delay in identification of the problems in the system such as leaks or surface pump and ESP issues. Also, most of the valves and chokes require to be operated manually. Due to the limited manpower, there are inadvertent delays in some surface operations and bringing online of the clusters.
Mitigation
The available manpower and resources are being utilized as best as possible to enhance the monitoring and operation of surface facilities. All the new WICs/SWICs and wells are equipped with real time monitoring and remote operational capabilities. The project for bringing the older facilities under the ambit of real time monitoring has been initiated and is expected to be completed in 3-4 years' time.

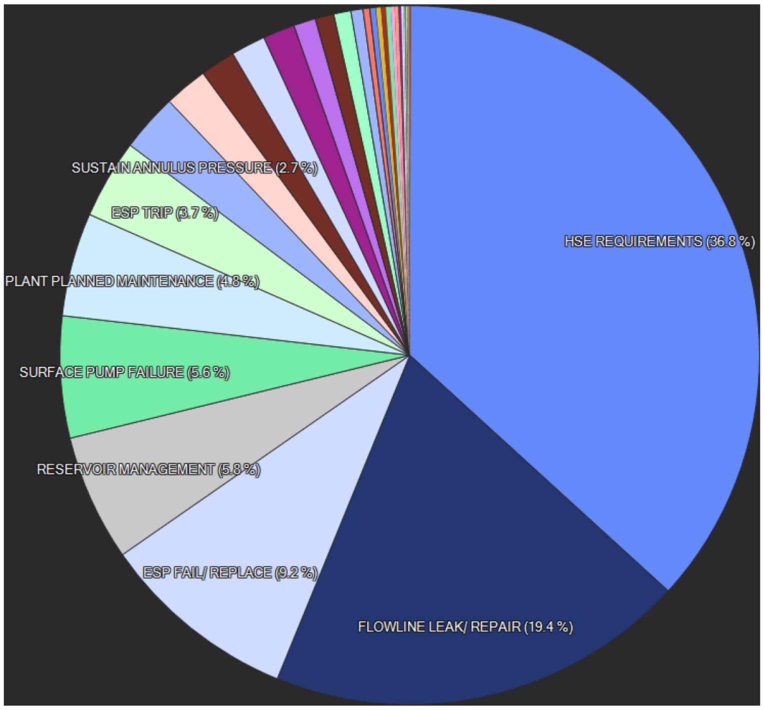

Figure 4—Causes for water injectors downtime

Reservoir Development

Field Development and Reservoir Management Issues

Water injection in reservoir-B initially commenced through peripheral injection. As more production-injection and reservoir data was acquired, higher field potential was identified through a modified field development plan involving line drive pattern with horizontal injectors and horizontal producers. The peripheral injectors have very low efficiency compared to the updip pattern injectors. The new injectors were connected to the available clusters wherever possible to reduce CAPEX for new clusters. This resulted in some injectors being extremely distant from the supply wells causing high pressure losses in the flowlines and, hence, reduced capacity. In some clusters, only the low efficiency peripheral injectors are connected from reservoir-B along with other reservoirs injectors which limit efficient utilization of the injection volumes. To keep the clusters operational, significant amounts of injection volumes are allocated to these peripheral injectors which reduced the overall injection support.

Because of the existing cluster system deficiencies, some regions in the reservoir-B have been facing lack of injection volumes despite presence of optimum number of injectors. As a result of this, these regions have developed pressure sinks. Few producers in these areas are unable to flow on self-due to low reservoir pressures amid rising water cut. Also, continuing production from these low pressure sink areas risks the development of a secondary gas cap.

Reservoir-B is a saturated reservoir with a large gas cap. As the field development strategy was revised in order to extract oil at higher withdrawal rates, the attic region was targeted to be developed with horizontal producers ad injectors in a line drive pattern. In order to limit the gas cap gas movement, a line of injectors along outer as well as inner gas cap was envisaged. The fence injection was planned to be commissioned with the sea-water injection project. To meet the increased production requirements, the withdrawal from attic region was increased substantially as other regions of the reservoir were already at the limit. This has led to rapid expansion of the gas cap which has resulted in the increasing GOR from the updip and attic producers. The increased gas production is constraining the existing production facilities which in turn is limiting the production from the field.

Mitigation Plan

The latest history matched simulation models were used to determine the injector-producer allocation factors and the injection efficiency for the reservoir-B. The injection efficiency is defined as the volume of oil in stock tank conditions supported by per barrel of water injected in stock tank condition by the injector. The study showed extremely low injection efficiencies for the peripheral injectors and the maximum efficiency for pattern injectors located near producers with zero or very low water cut. On the basis of this study several initiatives were proposed to the production optimization and operations teams and are currently under execution. These initiatives are described below.

Inter-Reservoir Injection Volume Transfer. A detailed exercise involving all the reservoir management teams in the field was carried out to determine if any additional injection is available in any of the reservoirs that can be spared for reservoir-B. The exercise was carried out on a cluster-by-cluster basis. A total of XX MBD volume was identified to be diverted to reservoir-B from other reservoirs.

Intra-Reservoir Injection Volume Transfer. Within reservoir-B, an initiative was undertaken to identify low efficiency injectors whose injection can be diverted to other high efficiency injectors without compromising on the reservoir health (no injection reduction in the pressure sink areas). The main emphasis was on identifying the active peripheral injectors that could be shut and there injection diverted to high efficiency pattern injectors. ~ XX MBD injection volume was identified for diversion among reservoir-B injectors.

Cluster Interconnections. A large portion of the inter and intra-reservoir volume transfer identified in above steps could not be materialized due to high efficiency reservoir-B wells not being hooked up with the cluster with additional volumes available. Hence, interconnection of clusters was proposed to efficiently divert the additional volumes in a cluster to another cluster with more requirement without affecting the cluster operation.

Re-Distribution of Injectors Among Clusters. Any injector, when it comes online, is connected to a cluster which is nearby and with spare capacity. Once hooked to a cluster, these connections are not readily changed as they are costly to change and require a lot of effort. Over time, as new injectors and clusters were commissioned, some of the connections are inefficient due to being extremely long. As a part of optimization, these inefficient connections were evaluated for replacement by connecting injectors with a nearby cluster which reduces the flowline pressure losses and hence improves injection capacity. Of the connections evaluated, some were taken up for replacement while others were rejected due to the overall cost not being supported by the oil volumes gained.

Objective Oriented Reservoir Management Program. A very comprehensive, objective-oriented reservoir management program has been implemented which is proactive in nature. As the field and, hence, the reservoirs are suffering from a lack of injection volumes, significant amount of Bottom Hole Close In Pressure (BHCIP) measurements are scheduled and conducted for identifying the areas of concern and the efficacy of the measures undertaken for remedying the low-pressure area creation. On basis of BHCIP measurements, Isobar maps are prepared which help in identifying areas that require more injection - this helps to identify the areas expedite drilling of the new injectors and SWICs commissioning.

The existing injectors are regularly surveyed for injectivity index and presence of skin through regular multi rate Bottom Hole Injection Pressure (BHIP) and Pressure Fall Off (PFO) surveys. These surveys complement the routine monitoring methods of Hall Plot and ABC plots to identify any wells that might be showing a gradual or sudden decline in injectivity and hence, schedule a suitable stimulation treatment based on information gathered from sources like scale or sludge sample recovered from the well, cluster and wellhead injection water analysis, etc. To avoid shut down of injectors for installation/retrieval of gauges, surface gauges are being employed.

Injection Logs and Pulsed Neutron logs have been used heavily to identify the fluid front movement and the conformance in the injectors to help identify any unswept regions as well as any issues in the wellbore affecting injection. This also helps to identify the stimulation candidates for conformance improvement and improving the effective sweep and injection capacity.

Modifications to Well Design. The field was initially developed using vertical producers and injectors. With more reservoir data and technological advancement, the field development envisaged horizontal producers and injectors in a line drive pattern. The lateral well lengths were limited to around 3000 ft per well. With increasing production from the wells, this resulted in higher drawdowns at the sand face. This results in faster water and gas dropdown from the upper zone and faster movement of water to the producers resulting in more unswept oil. Also, due to well length limited at ~3000 ft, a large of wells needed to be drilled which would have created a lot of congestion in the surface facilities. Hence, to overcome these issues, extended reach (ER) and maximum reservoir contact (MRC) wells were planned. These long lateral wells help in decongesting the surface facilities, reduce the drilling time and also reduce the drawdown for the same amount of oil production/ water injection resulting in a more even sweep. Also, the reduction in well count and increased well lengths help to cover a larger area with a single SWIC/WIC.

The wells were initially completed as openhole completions. This resulted in variable pressure differential along the wellbore with maximum at the heel and minimum at the toe. Also, the stimulation could not be carried out at high pumping rates resulting in lower stimulated area. Also, as the well length increased, issues were encountered in accessing the complete wellbore. To overcome these issues, wells were planned with lower completions incorporating Limited Entry Liners (LEL). The LEL lower completion is designed for each well to evenly distribute the pressure differential along the wellbore. This also allows the use of frac pumps for pumping for stimulation at very high rates resulting in a much larger stimulated area around the wellbore (Mogensen 2023). The presence of lower completions helps in improving the wellbore accessibility for interventions using CT or tractors.

As part of the field digitalization program, many wells are being completed with permanent downhole gauges (PDHG) and distributed temperature/acoustic sensors (DTS/DAS). These will enable continuous pressure, temperature and noise measurement that will help determine the well performance and detect any issues without requiring any interventions.

The changes in well design will also be beneficial post changeover to sea-water injection.

Conclusions

Lessons From Case Study

The following broad lessons can be derived from the case study of the giant carbonate reservoir's legacy injection system which has been in operation for over half a century.

1. Over the course of field development, the field development plan will change from the initial plan, In most cases, it will be dramatically different from the initial plan and might require significant overhauling of the surface facilities. In the current case study, the injection system will be completely overhauled to enable sea-water injection through a central facility from the current cluster-based injection system.
2. Overhauling surface facilities will take significant time and CAPEX investment. During this time, issues from the legacy system in place will linger on. The issues with the existing cluster-based injection system have been highlighted in the above sections briefly.
3. The impact from these issues cannot be eliminated but can be minimized through a multi-pronged strategy involving all the teams working individually as well as together identifying the major issues and come up with improvised solutions. Several such solutions for overcoming the legacy injection system in the reservoir and field in this case study are described in the sections above. The solutions

should be technically as well as economically viable. It would be extremely preferable if the solutions implemented were not only a make-shift arrangement but would continue to add value even in the new system.

Most of the solutions such as modification of well design, field digitalization, flowline replacement, injection efficiency utilization for prioritizing injectors, etc. are long term measures which will add value even post cluster-based injection system is replaced. The others like additional SWICs, inter and intra-reservoir injection volumes transfer, interconnection of clusters, re-distribution of injectors among clusters are temporary fixes which will be in place till the cluster system is continued. These temporary solutions were evaluated for economic viability and technical difficulty in execution.

The solutions have helped in improving the pressures in the sink areas and limit the pressure decline in areas with very high withdrawal. These initiatives have also helped in meeting the field and reservoir production targets while also helping with significant cost savings.

Way Forward

The initiatives undertaken will continue for the time being until the sea-water injection system comes into place. While the temporary solutions will be discontinued, the more long-term solutions will be continued. The injector efficiency determined through streamline simulation will be extended to other reservoirs to better determine the value of water (injector efficiency) for all the injectors in the system irrespective of the reservoir. This will help with the re-distribution of injection volumes more effectively.

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